

EE 451: Communications Systems

Midterm Exam 2 — Study Guide

Exam Date: Tuesday, April 21, 2026 (L21) **Duration:** 75 minutes **Coverage:** Chapters 4-7, 9, 11 (Angle Modulation, Digital Modulation, Noise & Link Budgets)

Exam Format

This exam has two parts, both completed during the 75-minute class period:

Part A — Multiple Choice (20 points, ~15 minutes)

- **20 questions** at 1 point each on **Brightspace**
- Covers conceptual knowledge from the L11, L13, and L17 reading quizzes, plus topics from L15-L16 and L18-L20
- Questions are drawn randomly from the same pool as the **Practice Quiz** (see below)
- You may complete Part A and Part B in any order

Part B — Open-Ended Problems (80 points, ~55 minutes)

- **5 problems** at 16 points each on **paper**
 - **Closed book, closed notes** — a formula sheet is provided
 - You may annotate the formula sheet with your own notes (no worked problems)
 - Show all work for full credit
 - Problems parallel Homework 3 and Homework 4
 - You may use a scientific calculator
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Practice Quiz (Available on Brightspace)

A **Practice Quiz** is available on Brightspace containing questions from the exam MC pool.

- **Unlimited attempts** — take it as many times as you want
 - **Your highest score counts as a homework grade**
 - Questions cover FM/PM, FSK, BPSK, baseband digital, noise concepts, and more
 - The best way to prepare for Part A is to take the practice quiz until you can consistently score well
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What to Study

For Part A (Multiple Choice)

The MC questions test conceptual understanding and quick recall. Focus on:

From L11 Reading (Ch 4.1-4.5): - FM vs. PM distinction - Frequency deviation and modulation index β - NBFM vs. WBFM classification - Carson's rule for FM bandwidth - Bessel functions and FM spectrum

From L13 Reading (Ch 4.6-4.8, 7.3-7.4): - FM generation methods (direct, indirect) - FM demodulation (discriminator, PLL) - Binary FSK fundamentals (modulation index h , orthogonality) - BPSK signal expressions and coherent detection

From L17 Reading (Ch 5-6): - Intersymbol interference (ISI) and its causes - Eye diagram interpretation - Nyquist criterion for zero ISI - Raised cosine filter and roll-off factor α

Additional MC topics from L15-L16, L18-L20: - Thermal noise ($P_n = kTB$), AWGN, SNR concepts - Noise figure, Friis cascaded noise figure - Link budget concepts (EIRP, path loss, link margin) - OFDM basics (subcarriers, cyclic prefix, advantages) - Spread spectrum (DSSS, FHSS, processing gain) - CDMA and Walsh codes - M-ary modulation (QPSK, QAM), spectral efficiency, EVM

Preparation strategy: Take the practice quiz on Brightspace repeatedly. Review any questions you get wrong.

For Part B (Open-Ended Problems)

The open-ended problems test your ability to set up and solve calculations. Problems are similar in style to your homework assignments. You should be comfortable with the following topics:

FM and PM Analysis - FM frequency deviation $\Delta f = k_f \cdot \max |m(t)|$ and modulation index $\beta = \Delta f / f_m$ - Carson's rule: $B_{\text{FM}} = 2(\Delta f + f_m)$ - PM phase deviation $\Delta \phi = k_p \cdot \max |m(t)|$ - PM frequency deviation: $\Delta f_{\text{PM}} = k_p \cdot A_m \cdot f_m$ (sinusoidal message) - FM vs. PM: frequency deviation dependence on message frequency

FSK and MSK - FSK modulation index $h = \Delta f \cdot T_b$ - MSK: $h = 0.5$, minimum frequency separation for orthogonality - FSK bandwidth estimation - Coherent vs. non-coherent FSK detection - Advantages of continuous-phase FSK (MSK)

BPSK, QPSK, and M-ary Modulation - BPSK and QPSK signal expressions - Constellation diagrams and Gray coding - Symbol rate $R_s = R_b / \log_2(M)$ - Null-to-null bandwidth for PSK - Spectral efficiency $\eta = R_b / B$ (bits/s/Hz) - Trade-off between spectral efficiency and power efficiency

Noise and Cascaded Receivers - Thermal noise power: $P_n = kTB$ - dB/linear conversions for gain and noise figure - Friis cascaded noise figure formula - Equivalent noise temperature $T_e = (F - 1) \cdot T_0$ - System noise temperature $T_{\text{sys}} = T_{\text{ant}} + T_e$ - SNR calculation from signal power and noise power

Link Budgets - EIRP = $P_t + G_t$ (dB) - Free-space path loss: $L_p = 20 \log_{10}(d) + 20 \log_{10}(f) + 32.45$ - Received power: $P_r = \text{EIRP} - L_p + G_r$ - Noise power: $N = -174 + 10 \log_{10}(B)$ (dBm) - Link margin = $\text{SNR} - \text{SNR}_{\text{req}}$

Preparation strategy: Redo Homework 3 and Homework 4 problems from scratch. If you can solve the homework without looking at solutions, you are well-prepared for Part B.

Essential Formulas

A formula sheet is provided with the exam and distributed ahead of time via the student GitHub repo. It includes FM/PM, digital modulation, noise, and link budget formulas. You do not need to

memorize these. However, you should know **how to use** each formula — practice applying them on homework problems.

You may annotate the formula sheet with your own notes, definitions, or reminders. However, **worked problems are not allowed** — formula sheets containing worked problems will be replaced with a clean copy at the start of the exam.

Key formulas to be comfortable with

FM/PM:

$f_i(t) = f_c + k_f \cdot m(t)$	[FM instantaneous frequency]
$\Delta f = k_f \cdot \max m(t) $	[FM frequency deviation]
$= \Delta f / f_m$	[FM modulation index]
$B_{FM} = 2(\Delta f + f_m)$	[Carson's rule]
$\Delta f_{PM} = k_p \cdot A_m \cdot f_m$	[PM frequency deviation, sinusoidal]

Digital Modulation:

$T_b = 1/R_b$	[bit duration]
$h = \Delta f \cdot T_b$	[FSK modulation index]
$\Delta f_{min} = 1/(2T_b)$	[coherent FSK orthogonality]
$R_s = R_b / \log(M)$	[symbol rate]
$B_{null} = 2R_b$ (BPSK), R_b (QPSK)	[null-to-null bandwidth]
$= R_b / B$	[spectral efficiency]

Noise:

$P_n = kTB$	[thermal noise]
$F_{total} = F + (F-1)/G + (F-1)/(GG)$	[Friis formula]
$T_e = (F - 1) \cdot T$	[equivalent noise temp]
$T_{sys} = T_{ant} + T_e$	[system noise temp]

Link Budget:

$EIRP = P_t + G_t$ (dB)	
$L_p = 20\log(d) + 20\log(f) + 32.45$	[d in km, f in MHz]
$P_r = EIRP - L_p + G_r$ (dBm)	
$N = -174 + 10\log(B)$ (dBm)	[B in Hz]
$Margin = SNR - SNR_{req}$	

Time Management (75 minutes)

You may complete Part A and Part B in any order. Here are suggested time targets:

Phase	Time	What to Do
Part A: Multiple Choice	~15 min	Answer 20 MC questions on Brightspace
Problem 1: FM/PM	~10 min	Deviation, modulation index, Carson's, FM vs PM
Problem 2: FSK/MSK	~10 min	Bit duration, orthogonality, bandwidth, MSK advantages

Phase	Time	What to Do
Problem 3: BPSK/QPSK/QAM	~10 min	Constellation, Gray coding, spectral efficiency
Problem 4: Noise	~10 min	Friis formula, noise temperature, SNR
Problem 5: Link Budget	~10 min	EIRP, path loss, received power, margin
Review	~10 min	Check answers, verify units and dB conversions
Total	75 min	

Common Mistakes to Avoid

FM/PM: - Confusing frequency deviation (Δf , in Hz) with modulation index (β , dimensionless)
- Forgetting to multiply Carson's rule by 2: $B_{\text{FM}} = 2(\Delta f + f_m)$ - In PM, forgetting that frequency deviation depends on f_m : $\Delta f_{\text{PM}} = k_p \cdot A_m \cdot f_m$

FSK/MSK: - Confusing frequency separation $\Delta f = |f_1 - f_0|$ with the distance from center ($\Delta f/2$)
- Forgetting the orthogonality condition: $\Delta f_{\text{min}} = 1/(2T_b)$, not $1/T_b$

BPSK/QPSK/QAM: - BPSK null-to-null bandwidth is $2R_b$, but QPSK is $2R_s = R_b$ (half the bandwidth for the same bit rate) - Forgetting that QPSK carries 2 bits/symbol, so $R_s = R_b/2$ - Gray coding: adjacent symbols differ by **one** bit, not two

Noise: - Mixing up linear and dB values in the Friis formula — the formula uses **linear** gains and noise figures - Forgetting to convert noise figure from dB to linear before applying Friis - Using $T_0 = 290$ K (the reference temperature), not T_{ant} , in the $T_e = (F - 1) \cdot T_0$ formula

Link Budget: - Path loss formula units: d in **km**, f in **MHz** — wrong units give wrong answers
- Forgetting that path loss is **subtracted** in dB: $P_r = \text{EIRP} - L_p + G_r$ - Doubling distance adds $20 \log_{10}(2) = 6.02$ dB of path loss (not 3 dB)

Self-Assessment Checklist

Before the exam, make sure you can:

FM and PM: - [] Calculate Δf , β , and classify as NBFM or WBFM - [] Apply Carson's rule and compare FM bandwidth to AM bandwidth - [] Calculate PM phase and frequency deviation - [] Explain the FM vs. PM difference in frequency deviation behavior

FSK and MSK: - [] Calculate FSK modulation index h and verify orthogonality - [] Write FSK signal expressions for both bit values - [] Estimate FSK/MSK bandwidth - [] Explain MSK advantages (continuous phase, constant envelope)

BPSK, QPSK, and QAM: - [] Write BPSK and QPSK signal expressions - [] Draw constellation diagrams with Gray coding - [] Calculate symbol rate for any M-ary scheme - [] Compute and compare spectral efficiencies

Noise and Cascaded Systems: - [] Convert between dB and linear for gain and noise figure - [] Apply the Friis cascaded noise figure formula - [] Calculate equivalent and system noise

temperatures - [] Calculate noise power and SNR in dB

Link Budgets: - [] Calculate EIRP - [] Calculate free-space path loss from distance and frequency
- [] Complete a full link budget (EIRP $\rightarrow$ path loss $\rightarrow$ received power $\rightarrow$ SNR $\rightarrow$ margin) - []
Explain the effect of doubling distance on path loss

Study Resources

Homework (Most Important!)

- **Homework 3:** FM/PM analysis, Carson's rule, FSK modulation index, BPSK expressions
- **Homework 4:** M-ary PSK/QAM, spectral efficiency, EVM calculations, WiFi/cellular examples
- Redo these problems from scratch — don't just re-read solutions

Practice Quiz

- Available on Brightspace — take it multiple times
- Covers all MC topics; highest score counts as homework

Lecture Notebooks

- **L10:** Angle modulation introduction (FM/PM)
- **L11:** FM spectrum, Bessel functions, Carson's rule, FSK
- **L12:** FSK demodulation, PLL, discriminator
- **L13:** BPSK modulation, constellation diagrams
- **L15:** Noise fundamentals, AWGN, SNR, noise figure
- **L17:** ISI, eye diagrams, raised cosine filter
- **L19:** Spread spectrum, CDMA
- **L20:** M-ary modulation, QAM, EVM, FT8

Textbook

- Chapter 4.1-4.5: FM/PM theory and modulation
 - Chapter 4.6-4.8: FM generation and demodulation
 - Chapter 5-6: Baseband digital transmission
 - Chapter 7.3-7.4: FSK, BPSK
 - Chapter 7.5-7.7: QPSK, QAM, spectral efficiency
 - Chapter 9: Noise fundamentals
 - Chapter 11.4-11.7: Link budgets
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Good luck!